Literature highlight report

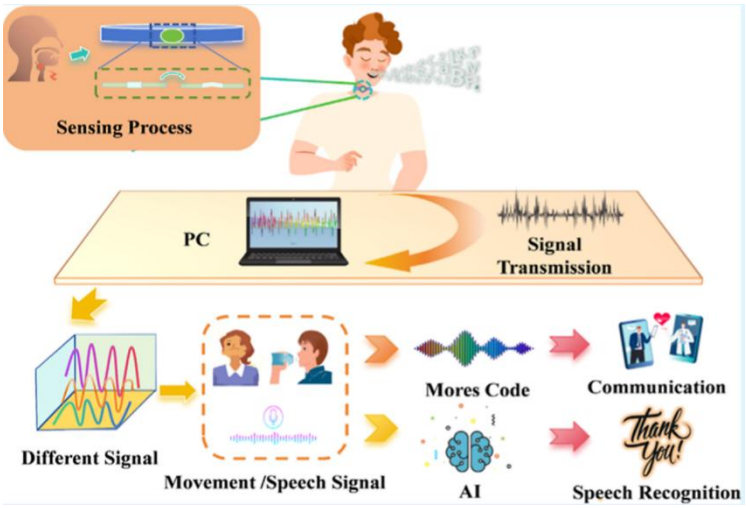
1. Wearable Photonic Artificial Throat for Silent Communication and Speech Recognition

This paper presents a photonic artificial throat which is a soft, wearable optical sensor placed on the neck that captures minute vibrations and muscle movements involved while speaking. Compare to microphone system, this device directly detect the source of sound produced, which means these signals are robust even during whispering, silent speech, or high-noise environments. The authors integrate the photonic sensor with a deep-learning-based classifier, achieving recognition of spoken and silent commands with high accuracy despite extremely low acoustic energy.

One of the two key innovations lies in the design of Photonic Artificial Throat which features a sandwich-structured optical fiber, the fiber is packaged in soft elastomer and fitted with a stress-concentrating ring to boost sensitivity and the light source and photo-detector(PD) are peripheral, muscle movement bends the optical fibre and PD collect light intensity changes to put into deep learning models.

Then it leads to the second innovation: using BKA-DNN based-algorithm to train the model which is a better way to deal with small amount of sample. BKA (Black-Kite Algorithm) is a meta-heuristic optimization algorithm, using weights and biases to optimize the 7 variables for DNN(Deep Neural Network): mean, variance, entropy, skewness, kurtosis, frequency, RMS.

This paper is highly relevant to my project because it give an example for using optical or photodiode-based sensing to reconstruct phonemes by using deep learning. Its methodology just parallels our objective of designing a photodiode or camera-based phoneme recognition model. It also provides engineering guidance on deep learning model choices.



2. Lip Reading Sentences Using Deep Learning With Only Visual Cues

The authors build a deep learning framework for sentence-level visual speech recognition using only lip movement videos. The study use the existing LRS2 “Lip Reading Sentences in the Wild” corpus, which contains tens of thousands of spoken sentences extracted from BBC broadcasts. This dataset enables training and evaluation under realistic, unconstrained conditions.

The basic pipeline for their work is using 3D CNN + 2D ResNet to extract Spatio-temporal characteristic and spatial characteristic respectively for video, then using Transformer encoder–decoder to classify viseme which can be reconstructed to words and sentences by GPT as a Decoder-Only Transformer. This viseme-to-word decoding strategy significantly improves speech recognition performance compared to previous end-to-end CNN+LSTM method or character-based transformer baselines.

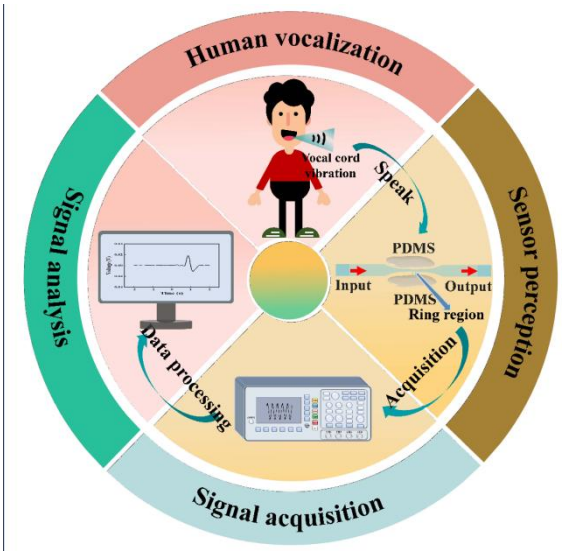
A key contribution of the work is that they set the smallest unit for classification as viseme for lip reading then use GPT perplexity dealing with linguistic combination to reconstruct sentences. The idea of making the complex problem into two separate parts of visual task and lingual task is why the makes the breakthrough.

This study is highly relevant to my project, as it provides a validated pipeline for camera-based images/video input for speech recognition. The paper offers valuable methodological insights into dividing our linguistic and visual tasks and how to choose the smallest linguistic unit. Also for the model choice, this study gives me deeper understanding of using different deep-learning models to fit their own tasks.

3. Flexible wearable vibration optical sensor for voice recognition enhanced property by a microfiber interferometer

This paper gives a framework of making neck skin vibration while speaking into predicted spoken word or command. The authors capture the skin vibration by flexible optical vibration sensor (PDMS waveguide) to modulates light intensity. Then they did analog signal conditioning containing the steps of amplification, high-pass filter for eliminating DC and baseline drift, low-pass filter for high frequency noises (>3 kHz) and normalization, processes it into acoustic features including to FFT spectrum, Mel-frequency cepstral coefficients (MFCC), Temporal envelope (amplitude modulation) and Spectro-temporal features, and performs speech recognition using deep learning models such as CNNs, LSTMs, and Transformers.

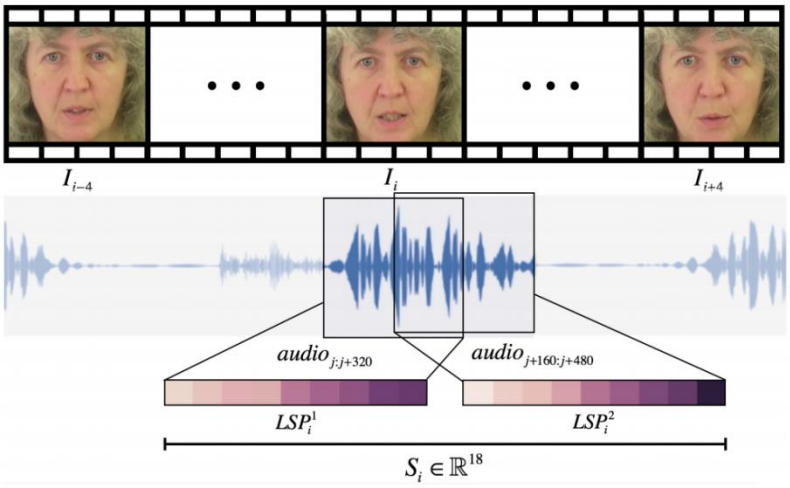
The most important part of the paper that I think I can learn from is the signal processing part, the authors use Photodetector to convert optical data to electrical signal then use FFT to perform spectrum without using deep learning models which reduce the complexity of the whole project. After getting those feature spectrum, the deep learning part become speech recognition from Mel spectrogram. This paper didn't mention the model part that much, they just use neural network as evaluation tools.



4. Vid2speech: Speech reconstruction from silent video

This paper proposes an end-to-end convolutional neural network (CNN) model for directly reconstructing the corresponding speech output from silent video. The system's input is a short-term grayscale facial video, and the output is the line spectral pair (LSP) parameters got from 8th-order linear prediction coding (LPC) analysis. The network used a structure of 3×3 receptive fields similar to VGG (Visual Geometry Group), which will automatically extracting spatio-temporal features from the entire face, it perform better than lip reading as the facial information is more completed, especially in distinguishing visually similar pronunciation vowels. The LSP sequence generated by the model is then converted into a waveform through a vocoder, resulting in clear and understandable speech. Notably, this method can reconstruct sentences not present in the data library, demonstrating a generalization ability and showing potential application value for silent speech synthesis.

This paper is valuable for me because it explain well about how did they using video as input for CNN model. The pre-processing stage of video is clear in the paper which conclude downsampling frequency, frame rate, overlap time and scale of video as reference for our project. And for training model, the paper mentioned their choice of activation function: Leaky ReLU and tanh function, the Adam optimizer, and also the method to avoid over-fitting like using Dropout and terminated training process the at loss stopped decreasing. The above training method can be used to train our infrared camera images.



References

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